

Lorentz covariant expressions out of Weyl spinors

[1.] To show that ϵ is an invariant tensor under the action of $SL(2, \mathbb{C})$, we can use the transformation properties of left and left dual Weyl spinors.

In matrix formalism, we have:

$$\psi'_L = L\psi_L \quad \text{and} \quad \tilde{\psi}'_L = \psi'^L_L \epsilon' = \tilde{\psi}_L L^{-1},$$

where $L \in SL(2, \mathbb{C})$ and $\det L = 1$. Overall we get

$$\begin{aligned} \tilde{\psi}'_L &= \psi'^L_L \epsilon' = (L\psi_L)^T \epsilon' = \psi^T_L L^T \epsilon' \\ &= \tilde{\psi}_L L^{-1} = \psi^T_L \epsilon L^{-1}. \end{aligned}$$

Since this relation must be valid for all ψ_L , we obtain

$$\epsilon = L^T \epsilon' L = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} ac - ca & ad - cb \\ bc - ad & bd - db \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

where we have used $\det L = ad - cb = 1$, and therefore we have proven that ϵ is invariant.

In index notation, we have

$$\psi'^{\mu} = L^{\mu}_{\nu} \psi^{\nu} \quad \text{and} \quad \psi'_{\nu} = \psi'^{\mu} \epsilon'_{\mu\nu} = L_{\nu}^{\mu} \psi_{\mu},$$

where L_{ν}^{μ} is the inverse matrix of L^{μ}_{ν} , i.e. $L^{\mu}_{\nu} L^{\nu}_{\kappa} = \delta^{\mu}_{\kappa}$. Similarly, we get

$$\begin{aligned} \psi'_{\nu} &= \psi'^{\mu} \epsilon'_{\mu\nu} = L^{\mu}_{\kappa} \psi^{\kappa} \epsilon'_{\mu\nu} \\ &= L_{\nu}^{\mu} \psi_{\mu} = L_{\nu}^{\mu} \psi^{\kappa} \epsilon_{\kappa\mu}, \end{aligned}$$

which again must be true for all ψ^{κ} ; therefore,

$$\epsilon_{\kappa\lambda} = L^{\mu}_{\kappa} L^{\nu}_{\lambda} \epsilon'_{\mu\nu}.$$

For $\kappa = \lambda$, we use the symmetry of μ, ν in L and the antisymmetry of μ, ν in ϵ' , obtaining $\epsilon_{\kappa\kappa} = 0$. For $\kappa \neq \lambda$, we can choose, for example,

$$\epsilon_{12} = L^{\mu}_1 L^{\nu}_2 \epsilon'_{\mu\nu},$$

which is the defining relationship for the determinant of matrix L , which equals one. For ϵ_{21} , we get the desired result by again using the antisymmetry of ϵ' .

In addition, if we use the transformation formula:

$$\epsilon_{\kappa\lambda} = L^\mu{}_\kappa L^\nu{}_\lambda \epsilon'_{\mu\nu},$$

as the defining formula for the Lorentz transformation of the ϵ tensor, it is natural because ϵ is a 2-form. However, its invariance under Lorentz transformations specifically follows from the special property of the L matrices having unit determinant ($\det L = 1$). This property allows ϵ to act as a metric for raising and lowering spinor indices.

Using this relation, we can work backwards to derive the Lorentz transformation properties of other objects, starting, for example, from the transformation properties of ψ_L , as requested in the '0.' exercise.

2.

Rotations

A rotation around axis i by an angle φ is represented as:

$$L_i^R(\varphi) = e^{i\frac{\varphi}{2}\sigma_i}.$$

The explicit forms of the 2×2 rotation matrices, derived in the previous homework, are:

$$L_x^R(\varphi) = \begin{pmatrix} \cos \frac{\varphi}{2} & i \sin \frac{\varphi}{2} \\ i \sin \frac{\varphi}{2} & \cos \frac{\varphi}{2} \end{pmatrix},$$

$$L_y^R(\varphi) = \begin{pmatrix} \cos \frac{\varphi}{2} & \sin \frac{\varphi}{2} \\ -\sin \frac{\varphi}{2} & \cos \frac{\varphi}{2} \end{pmatrix},$$

$$L_z^R(\varphi) = \begin{pmatrix} e^{i\frac{\varphi}{2}} & 0 \\ 0 & e^{-i\frac{\varphi}{2}} \end{pmatrix}.$$

Symmetry Rotation matrices are not symmetric in general. For example, $L_y^R(\varphi)$ (rotation around y -axis) satisfies $L_y^R(\varphi)^T \neq L_y^R(\varphi)$. Thus, rotation matrices are generally *not symmetric*.

Orthogonality Rotation matrices in $SL(2, \mathbb{C})$ are not orthogonal in the standard sense because they do not preserve the real inner product. For example, $L_z^R(\varphi)$ involves complex exponentials.

Unitarity All rotation matrices are unitary because they are generated by anti-Hermitian operators ($i\sigma_j$). For any rotation matrix L :

$$L^\dagger L = I.$$

Group Formation To verify that rotations form a group, consider two infinitesimal rotations around axes j and k :

$$L_j^R(\varphi) = I + i\frac{\varphi}{2}\sigma_j - \frac{\varphi^2}{4}\sigma_j^2 + \mathcal{O}(\varphi^3),$$

$$L_k^R(\varphi) = I + i\frac{\varphi}{2}\sigma_k - \frac{\varphi^2}{4}\sigma_k^2 + \mathcal{O}(\varphi^3).$$

Their product, retaining terms up to second order, is:

$$L_j^R(\varphi)L_k^R(\varphi) = I + i\frac{\varphi}{2}(\sigma_j + \sigma_k) - \frac{\varphi^2}{4}(\sigma_j^2 + \sigma_k^2) - \frac{\varphi^2}{4}\sigma_j\sigma_k + \mathcal{O}(\varphi^3).$$

Using the Pauli matrix identity:

$$\sigma_j\sigma_k = \delta_{jk}I + i\epsilon_{jkl}\sigma_l,$$

the term $-\frac{\varphi^2}{4}\sigma_j\sigma_k$ becomes:

$$-\frac{\varphi^2}{4}(\delta_{jk}I + i\epsilon_{jkl}\sigma_l).$$

Substituting back, we obtain:

$$L_j^R(\varphi)L_k^R(\varphi) = I + \frac{i}{2}\vec{\varphi}' \cdot \vec{\sigma} - \frac{1}{8}(\vec{\varphi}' \cdot \vec{\sigma})^2 + \mathcal{O}(\varphi^3),$$

where the new effective rotation vector is:

$$\vec{\varphi}' \cdot \vec{\sigma} = \varphi\sigma_j + \varphi\sigma_k + i\frac{\varphi^2}{2}\epsilon_{jkl}\sigma_l.$$

This result shows that the product of two rotations is equivalent to a single rotation with a new effective axis and angle. Rotations in $SL(2, \mathbb{C})$ therefore form a group $SU(2)$, which is the double cover of the $SO(3)$ group.

Boosts

A boost in the direction i with rapidity u is represented as:

$$L_i^B(u) = e^{\frac{u}{2}\sigma_i}.$$

The explicit forms of the 2×2 boost matrices, derived in the previous homework, are:

$$\begin{aligned} L_x^B(u) &= \begin{pmatrix} \cosh \frac{u}{2} & \sinh \frac{u}{2} \\ \sinh \frac{u}{2} & \cosh \frac{u}{2} \end{pmatrix}, \\ L_y^B(u) &= \begin{pmatrix} \cosh \frac{u}{2} & -i \sinh \frac{u}{2} \\ i \sinh \frac{u}{2} & \cosh \frac{u}{2} \end{pmatrix}, \\ L_z^B(u) &= \begin{pmatrix} e^{\frac{u}{2}} & 0 \\ 0 & e^{-\frac{u}{2}} \end{pmatrix}. \end{aligned}$$

Symmetry Boosts are not symmetric in general. For instance, $L_y^B(u)$ satisfies $L_y^B(u)^T \neq L_y^B(u)$. Thus, boost matrices are generally *not symmetric*.

Orthogonality Boosts are not orthogonal transformations because they do not preserve the inner product:

$$L_z^B(u)^T L_z^B(u) \neq I.$$

Unitarity of Boosts Boosts are not unitary because they are generated by Hermitian operators (σ_i) without an imaginary factor i in their exponent. This results in matrices that do not satisfy the condition $L^\dagger L = I$. Instead, boost matrices modify the norm of spinors, which reflects their role in Lorentz transformations as non-unitary operations in Minkowski spacetime.

Group Formation Consider two infinitesimal boosts along directions j and k :

$$\begin{aligned} L_j^B(u) &= I + \frac{u}{2}\sigma_j + \frac{u^2}{4}\sigma_j^2 + \mathcal{O}(u^3), \\ L_k^B(u) &= I + \frac{u}{2}\sigma_k + \frac{u^2}{4}\sigma_k^2 + \mathcal{O}(u^3). \end{aligned}$$

Their product, retaining terms up to second order, is:

$$L_j^B(u)L_k^B(u) = I + \frac{u}{2}(\sigma_j + \sigma_k) + \frac{u^2}{4}(\delta_{jk}I + i\epsilon_{jkl}\sigma_l) + \mathcal{O}(u^3).$$

The term $i\epsilon_{jkl}\sigma_l$ corresponds to a rotation, which cannot be interpreted as part of a pure boost. This is the Wigner-Thomas rotation arising at second order. The remaining terms correspond to an effective boost in a new direction. Thus, boosts alone are not closed under composition and do not form a group.

3.

(i) $\psi_L^\dagger \psi_R$: For this type of object, we can use the general rules for Lorentz transformations, which hold for both rotations and boosts or any general transformation. We have:

$$\left(\psi_L^\dagger \psi_R\right)' = \psi_L'^\dagger \psi_R' = \psi_L^\dagger L^\dagger (L^{-1})^\dagger \psi_R = \psi_L^\dagger (L^{-1} L)^\dagger \psi_R = \psi_L^\dagger I^\dagger \psi_R = \psi_L^\dagger \psi_R.$$

We see that the studied object is invariant under Lorentz transformations, which corresponds to a covariant transformation for a scalar-type object. This reflects the fact that this object has "no index." For both $SO(3)$ and $SO(1,3)$, the studied object is a type $(0,0)$ tensor — a scalar.

Similarly, we can see that the object $\psi_R^\dagger \psi_L$ is of the same tensor type as the first studied object:

$$\left(\psi_R^\dagger \psi_L\right)' = \psi_R'^\dagger \psi_L' = \psi_R^\dagger L^{-1} L \psi_L = \psi_R^\dagger I \psi_L = \psi_R^\dagger \psi_L.$$

Knowing how the left and right chiral Weyl spinors transform under parity:

$$\psi_L \xrightarrow{\mathcal{P}} \psi_R, \quad \psi_R \xrightarrow{\mathcal{P}} \psi_L,$$

we see that the two derived objects map into each other:

$$\psi_L^\dagger \psi_R \xleftrightarrow{\mathcal{P}} \psi_R^\dagger \psi_L.$$

Therefore, we can form symmetric and antisymmetric combinations to construct eigenstates of parity with eigenvalues ± 1 :

$$\Psi_\pm = \psi_L^\dagger \psi_R \pm \psi_R^\dagger \psi_L.$$

(ii) $\psi_L^\dagger \psi_L$: Using the transformation formulae, we obtained in the second exercise

$$\left(\psi_L^\dagger \psi_L\right)' = \psi_L'^\dagger \psi_L' = \psi_L^\dagger L^\dagger L \psi_L,$$

we derive that only the matrices representing rotations are unitary, whereas matrices representing boosts are not. This means that $\psi_L^\dagger \psi_L$, an index-less object, is a scalar under rotations (transformations of the group $SO(3)$). However, since boost matrices are not unitary, the object does not transform covariantly (as a scalar) under general Lorentz transformations in $SO(1, 3)$. Therefore, $\psi_L^\dagger \psi_L$ is not a Lorentz scalar.

(iii) $\psi_L^T \epsilon \psi_L + \psi_L^\dagger \epsilon \psi_L^*$: In this case, we can use the transformation rules for the dual representations:

$$\begin{aligned} (\psi_L^T \epsilon \psi_L)' &= (\tilde{\psi}_L \psi_L)' = \tilde{\psi}_L' \psi_L' = \tilde{\psi}_L L^{-1} L \psi_L = \tilde{\psi}_L I \psi_L = \tilde{\psi}_L \psi_L = \psi_L^T \epsilon \psi_L, \\ (\psi_L^\dagger \epsilon \psi_L^*)' &= [(\tilde{\psi}_L)^* \psi_L^*]' = (\tilde{\psi}_L)^*{}' \psi_L^{*'} = (\tilde{\psi}_L)^* (L^{-1})^* L^* \psi_L^* = (\tilde{\psi}_L)^* (L^{-1} L)^* \psi_L^* = \psi_L^\dagger \epsilon \psi_L^*. \end{aligned}$$

Here we see, similarly to the first case, that this type of indexless object is completely invariant under the entire Lorentz group, i.e., both under boosts and rotations. This invariance arises because the terms involve the antisymmetric invariant tensor ϵ , which remains unchanged under Lorentz transformations. Therefore, the given expression is a type $(0, 0)$ tensor—a scalar—for both $SO(3)$ and $SO(1, 3)$.

(iv) $\psi_L^\dagger \bar{\sigma}_\mu \psi_L$: In this case, we can use the hint in the worked-out example and form an object that should be an invariant, assuming the studied object transforms covariantly:

$$v^\mu \psi_L^\dagger \bar{\sigma}_\mu \psi_L = \psi_L^\dagger \mathbf{v}_R \psi_L.$$

Now, we verify the invariance:

$$(\psi_L^\dagger \mathbf{v}_R \psi_L)' = \psi_L^{\dagger'} \mathbf{v}_R' \psi_L' = \psi_L^\dagger L^\dagger (L^\dagger)^{-1} \mathbf{v}_R L^{-1} L \psi_L = \psi_L^\dagger I \mathbf{v}_R I \psi_L = \psi_L^\dagger \mathbf{v}_R \psi_L,$$

which is indeed invariant. This shows that contracting the studied object with a vector produces a Lorentz-invariant scalar. Therefore, the studied object itself must be a type $(0, 1)$ tensor—a covector.

Using the relations for the effect of the parity operator:

$$\psi_L \xleftrightarrow{\mathcal{P}} \psi_R, \quad \mathbf{v}_L \xleftrightarrow{\mathcal{P}} \mathbf{v}_R,$$

we see that:

$$\psi_L^\dagger \mathbf{v}_R \psi_L \xleftrightarrow{\mathcal{P}} \psi_R^\dagger \mathbf{v}_L \psi_R.$$

This again corresponds to a $(0, 1)$ -type tensor—a covector. Thus, after stripping v^μ , we can form parity eigenstates by taking symmetric and antisymmetric combinations:

$$\Phi_\pm = \psi_L^\dagger \bar{\sigma}_\mu \psi_L \pm \psi_R^\dagger \sigma_\mu \psi_R.$$

The object $\psi_L^\dagger \bar{\sigma}_\mu \psi_L$ transforms as a covector (covariant rank-1 tensor) under the Lorentz group $SO(1, 3)$, corresponding to the representation $(0, 1)$. Under $SO(3)$ (rotations), it decomposes into two parts: a time-like scalar (under rotations) and a spatial 3-vector.

(v) $\psi_L^T \epsilon \sigma_\mu \psi_L$: Similarly to the previous case, we assume the studied object transforms covariantly. Thus:

$$v^\mu \psi_L^T \epsilon \sigma_\mu \psi_L = \tilde{\psi}_L \mathbf{v}_L \psi_L$$

should form a Lorentz-invariant scalar. To verify this, we compute:

$$\left(\tilde{\psi}_L \mathbf{v}_L \psi_L \right)' = \tilde{\psi}_L' \mathbf{v}_L' \psi_L' = \tilde{\psi}_L L^{-1} L \mathbf{v}_L L^\dagger L \psi_L = \tilde{\psi}_L \mathbf{v}_L L^\dagger L \psi_L.$$

From this computation, we see that the studied object is Lorentz-invariant only if the matrix L is unitary. Therefore, this object, like in the second case, transforms as a $(0, 1)$ -type tensor under rotations, i.e., transformations of the $SO(3)$ group. However, it does not transform as a proper tensor under general Lorentz transformations, since L is not unitary for boosts.